

Beyond traditional pruning

Overview

Deep Neural Networks can solve extremely challenging tasks thanks to complex stacks of (convolutional) layers with thousands of neurons, especially in solving computer vision-related tasks like image classification, object detection, or image segmentation. Their success comes from their ability to learn from examples, not requiring any specific expertise, and using very general learning strategies, based on loss minimization. However, many deep models share a common drawback: their growing complexity challenges the computational capability of embedded devices and poses questions about power consumption when deployed on the field.

Deep neural network compression is hence an important area of research and development in the field of machine learning. While it offers many benefits, it also comes with several challenges that researchers and practitioners need to address. Here are some of the key challenges for DNN compression. **Maintaining Performance** is one of the primary challenges in DNN compression, which is reducing the size and computational complexity of a model while preserving or minimizing the loss in performance. Achieving a balance between compression and performance is crucial, and it often requires careful optimization and tuning. **Pruning Without Over-pruning** aims to remove redundant connections and neurons from a neural network. However, excessive pruning can lead to a loss of important features and a drop in model performance. Determining the right amount of pruning is challenging. A different approach relies on **Knowledge Distillation**, transferring knowledge from a large, complex model (teacher) to a smaller, more efficient model (student). Balancing the size and performance of the student model while effectively distilling knowledge from the teacher model is a non-trivial task. Of course, **Hardware and Deployment Constraints** play an important role on resource-constrained devices such as mobile phones or embedded systems. Adapting compressed models to various hardware platforms and ensuring efficient execution is challenging. Compressed models may have a tendency to overfit or be less robust to noisy data compared to their larger counterparts. Ensuring that compressed models **generalize** well and are robust in real-world scenarios is a challenge. Finally, **transferability** is one little explored subject associated with compressed neural networks, which can be properly performed by having a certain degree of **interpretability**.

In this project, we will be mainly working on pruning. After the study and implementation of a benchmark with all the most popular and effective approaches (structured VS unstructured and beyond) with a neat evaluation of a range of devices for the pruned approaches (which can include embedded devices) with evaluations in terms of energy and memory consumption for each of the tested approaches, involving study and deployment of approaches related to Neural Tangent Kernel theory and Optimal Transport. This exploration will validate or raise

concerns/questions of known concepts around model compression, that can lead to the design of new and more effective approaches. The perfect candidate has a strong background in Python coding and puts in massive passion and effort. Depending on the final quality and quantity of the results achieved, a submission of a paper will be taken into consideration.

Objectives at a glance

- Study the most recent state-of-the-art approaches around pruning.
- Deeply understand forward and back-propagation mechanisms and how these impact algorithmic complexity for these.
- Replicate/reproduce state-of-the-art results, with partial or total re-implementation of the approaches.
- Study the feasibility of efficiency improvements (both from the perspectives of energy consumption and HW requirements), scaling from simple to more complex tasks, from simple convolutional neural networks to transformers, and eventually to Large Language Models.
- Deployment on real embedded devices.

Tutor

The main responsible for the project is **Enzo Tartaglione**, MdC for the Multimedia equipe enzo.tartaglione@telecom-paris.fr.

Pre-interview for this project is **mandatory** and can be agreed upon by email.

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